FORMAL SMOOTHNESS RESTRICTS TO OPEN SUBSCHEMES OF SOURCE AND BASE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3140-3145 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 3147-3161, the complete author proof (13 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The lifting definition of formal smoothness (definition-formally-smooth, same chapter) and basic facts on open immersions (Schemes); cited inputs with the diagram argument written in the retained span.

Source reference. Original lemma 02H0: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 3082–3093. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01HD: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 3161–3161. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Let $U \subset X$ and $V \subset S$ be open subschemes such that $f(U) \subset V$. If f is formally smooth, so is $f|_U : U \to V$.*

Proof. Consider a solid diagram

$$\begin{array}{ccc} U & \xleftarrow{a} & T \\ {\scriptstyle f|_U}\downarrow & \nearrow & \downarrow{\scriptstyle i} \\ V & \longleftarrow & T' \end{array}$$

as in Definition 02H0. If f is formally smooth, then there exists an S-morphism $a' : T' \to X$ such that $a'|_T = a$. Since the underlying sets of T and T' are the same we see that a' is a morphism into U (see Schemes, Section 01HD). And it clearly is a V-morphism as well. Hence the dotted arrow above as desired. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.